B.7 Further Functions

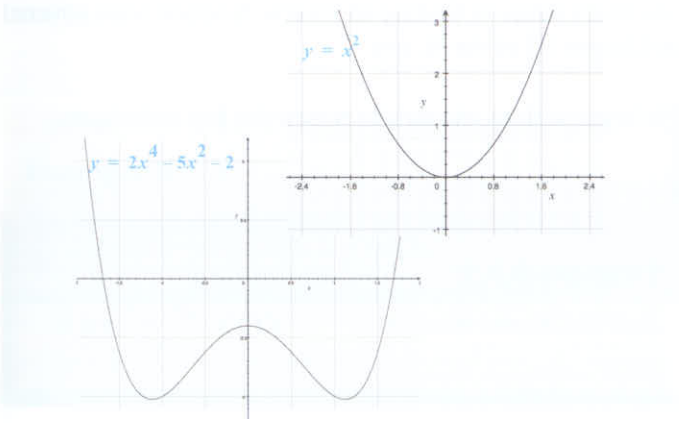
AHL 2.14
AHL 2.15

Odd and Even Functions

Functions can be classified into three categories based on the symmetries of their graphs.

Even functions

If a function has line symmetry about the y-axis, it is said to be even.



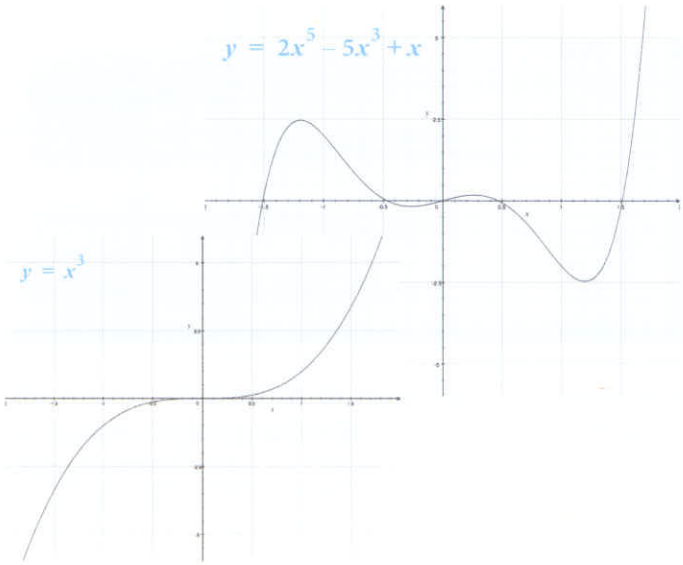
Formally, a function is even if $f(x) = f(-x)$ for all x in the domain.

The most obvious examples of even functions are the even polynomials, x^2, x^4... Other examples are $|x|$, $\cos(x)$.

Odd functions

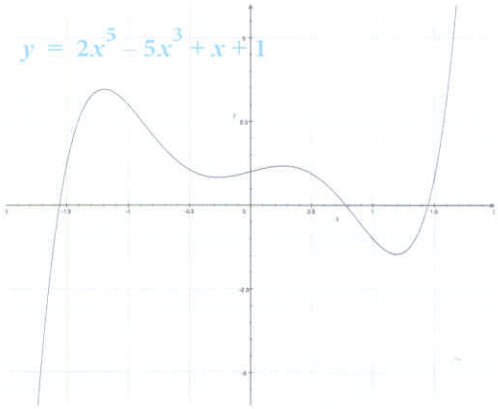
A function is odd if it has two-fold rotational symmetry about the origin. If the graph is 'pinned' at the origin and turned through 180°, it will fit back over the original graph.

Formally, a function is odd if $f(x) = -f(-x)$ for all x in the domain, e.g. x, x^3, $\sin(x)$.



The concept applies to other, non-polynomial functions. The cosine function is even, the sine function is odd and the logarithm function is neither.

Not every function is either odd or even. Most functions are neither.

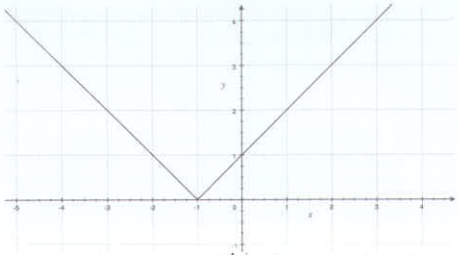


Example B.7.1

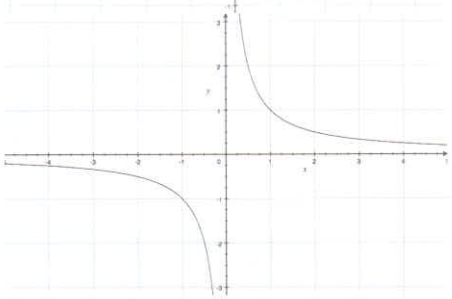
Classify the following functions as even, odd or neither.

a $f(x) = |x+1|$ b $f(x) = \frac{1}{x}$ c $f(x) = \left|\frac{1}{x}\right|$

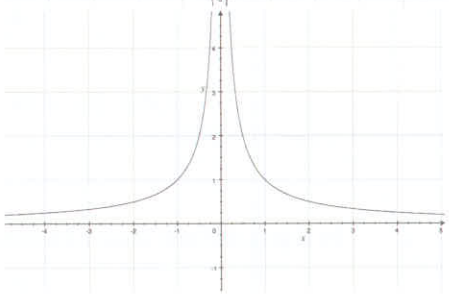
a Neither odd nor even.



b Odd



c Even



Exercise B.7.1

1. Classify the following functions as even, odd or neither.

a $y = 4$ b $y = |x^3|$

c $y = (x-1)^3$ d $y = \frac{x}{x-2}$

e $y = \ln(x^2)$ f $y = \frac{1}{x}$

g $y = \frac{x^2+2}{x^3+x}$ h $y = \frac{x}{x^3-x}$

i $y = \frac{\sin x}{\cos x} + x^3$

2. Prove that the product of two even functions is even.

3. Is it necessary for $f(0)=0$ for a function to be odd?

4. Explain why the composite of two odd functions is odd.

5. Prove that the quotient of two even functions is even.

6. A function has the full real line as its domain and is both odd and even. What is the rule for the function?

Identity and inverse functions

The main stages of finding an inverse function were covered in Chapter B2 of the SL text.

We begin with an example to review the key techniques.

Example B.7.2

Find the inverse function of $f(x)=\frac{2x-3}{5}, x \in \mathbb{R}$

The first step is to write this as a '$y=$' statement.

$$y=\frac{2x-3}{5}$$

Next, invert the function by exchanging x & y.

$$x=\frac{2y-3}{5}$$

Then, make y the subject:

$$5x=2y-3$$
$$2y=5x+3$$
$$y=\frac{5x+3}{2}$$
$$f^{-1}(x)=\frac{5x+3}{2}, x \in \mathbb{R}$$